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An experimental study of the performance of IEEE 802.11ad in smartphones

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ABSTRACT

We present the first extensive measurement study of the performance of the 60 GHz 802.11ad technology on commercial smartphones. We focus on five major perspectives: (i) TCP throughput and communication range in static scenarios, (ii) spatial and temporal variations, (iii) coverage, (iv) performance under realistic smartphone user mobility patterns, and (v) impact of human blockage. We also compare for the first time 802.11ad against its two main competitors in the 5 GHz band: the state-of-the-art 802.11ac and the upcoming 802.11ax, which also promises multi-Gbps data rates. Finally, we present a preliminary evaluation of the 802.11ad power consumption on a smartphone under Gbps data rates. Overall, our results show that, in spite of earlier concerns, 802.11ad has higher potential to address the needs of emerging bandwidth-intensive applications in smartphones than its 5 GHz counterparts. At the same time, our study identifies several open challenges and key research directions towards fully realizing this potential for future smartphone users.

1. Introduction

An emerging class of smartphone applications, such as mobile Augmented/Virtual reality (AR/VR), Miracast, and UHD video streaming, demand Gbps speeds from the underlying wireless network. The 14 GHz of unlicensed spectrum around 60 GHz have attracted ample attention from both academia and industry as one of the candidate solutions for providing the required Gbps data rates. The IEEE 802.11ad standard [1] governs the use of this spectrum and is touted as one of the main technologies for building the next generation of WLANs. The standard supports 2 GHz-wide channels and provides data rates of up to 6.7 Gbps, a multi-fold increase over legacy WiFi in the 5 GHz band. The upcoming 802.11ay standard, with support for channel bonding and MIMO, further pushes the envelope promising data rates of up to 100 Gbps.

Despite early concerns regarding the range and performance in indoor environments, 802.11ad-compliant APs and laptops have been released on the market over the past few years. However, until last year, there was no commercially available smartphone featuring an 802.11ad chipset. Researchers and smartphone vendors have always had several concerns with putting an 802.11ad chipset on a phone such as high power consumption [2,3], hardware resource usage [4], and antenna placement [5]. In fact, we believe that these concerns have been one of the main reasons that have slowed down a widespread adoption of 802.11ad as a mainstream WLAN technology. After a number of early failed attempts [6,7], ASUS finally released the ROG phone – the first smartphone featuring an integrated 802.11ad chipset – in March 2019[1] and the ROG II phone in September 2019.

Nonetheless, the performance of 802.11ad in smartphones remains largely unknown. Although recent studies using 60 GHz SDRs [8,9] or commercial-off-the-shelf (COTS) laptops, APs, or docks [2,10–13]) have shown that 802.11ad indeed works well indoors (the range is much longer than initially predicted and beamforming in COTS devices is fast enough to minimize periods of low connectivity), it is not clear if these findings can be extended to smartphones due to a number of reasons: (i) Antenna array placement becomes a non-trivial problem, as there are more than one ways of holding a smartphone. Since practical 802.11ad phased arrays cannot generate homogeneous beams across all directions due to the use of discrete phase-shifts on antenna elements [8,9,12,14], the antenna array placement along with the device orientation can have a significant impact on coverage. (ii) The small form factor limits the number of antenna elements one can pack into a smartphone, compared to a laptop or AP [15]. Smaller phased arrays generate fewer, wider beams with lower directivity gain, which can further reduce the device's coverage. (iii) The small form factor makes it much easier for the user to block the antenna with their hand or body [16]. (iv) The mobility patterns of smartphone users are much more diverse and unpredictable [17] compared to those of laptop users. (v) Hardware resources and power management policies can be a significant performance bottleneck in smartphones [4,18].

At the same time, legacy WiFi in the 5 GHz band has been evolving and offering increasingly higher data rates. 802.11ac, the prevalent WLAN standard today, already supports theoretical data rates up to 3.4 Gbps, although hardware constraints in smartphones limit the achievable throughput in practice to around 600 Mbps [4,18]. The latest 802.11ax standard (WiFi 6) [19] promises up to 6.9 Gbps, even

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[1] The phone was actually released in October 2018 but the use of the 802.11ad NIC was only enabled with a firmware update in March 2019.

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higher than the maximum 802.11ad rate. The first 802.11ax-equipped smartphones appeared on the market about one year ago. 802.11ax is often viewed as a strong competitor to 802.11ad, as it offers backward compatibility with and higher data rates than 802.11ac and, at the same time, longer range than 802.11ad and resilience to blockage and mobility. Nonetheless, to our best knowledge, no work has studied experimentally the performance of 802.11ax and compared it against 802.11ad.

In this paper, we fill this gap by performing the *first* extensive experimental study of the performance of 802.11ad on commercial smartphones. We focus our study on performance aspects affected by unique smartphone features, e.g., antenna array placement or user mobility patterns, and compare the performance against that achieved by 802.11ad laptops in previous studies. Our goals are two-fold: (i) to understand the performance of 802.11ad in smartphones and (ii) to compare 802.11ad against its two main competitors — 802.11ac and, for *first time*, 802.11ax, on mobile devices. We also perform a preliminary evaluation of the 802.11ad power consumption under Gbps data rates. We plan to make our measurement dataset publicly available. Our contributions and findings are summarized as follows:

Range, performance in static settings (Section 4): We conduct an extensive study of the performance of 802.11ad in a smartphone and compare it against (i) the performance of 802.11ad in a laptop and (ii) the performance of 5 GHz technologies in smartphones. The range of an 802.11ad AP-phone link is lower compared to that of an AP-laptop link and the uplink throughput is lower than the downlink, due to the poor performance of the beamforming protocol on the phone. The orientation at which the phone is held has a great impact on the 802.11ad performance. Additionally, the performance exhibits large temporal and spatial variations. Nonetheless, Gbps speeds are still feasible at distances of several tens of ft. On the other hand, even though 802.11ax is marketed as a Gbps technology, the first generation of 802.11ax phones does not support Gbps throughputs, *making 802.11ad the only real Gbps WLAN technology on today's smartphones*.

Coverage (Section 5): We find that the 802.11ad coverage with a phone is lower than with a laptop and much worse compared to 5 GHz WiFi. Additionally, the 802.11ad coverage is affected significantly by the device orientation, location, and AP-client angular separation. In contrast, 802.11ac and 802.11ax offer almost uniform coverage.

Impact of mobility (Section 6): We evaluate for first time the performance of 802.11ad under typical smartphone user mobility patterns. We find that the poor performance of the 802.11ad beamforming protocol and its interaction with rate adaptation (RA) results in very large throughput variations during motion, and self-blockage from the user-body can severely degrade the performance, calling for motion-aware RA and beamforming algorithms. On the other hand, keeping the phone in the pocket, surprisingly, does not block the 60 GHz signal between the phone and the AP.

Impact of blockage (Section 7): We explore the impact of various types of human blockage on the performance of an 802.11ad AP-phone link. We show that mobile human blockage has limited impact on the performance of an AP-phone link. In contrast, quasi-static human blockage has a severe impact on the performance of AP-phone links (much worse compared to AP-laptop links), which deteriorates as the blockage moves closer to the phone's location.

Power consumption (Section 8): We perform the first experimental study of the power consumption of 802.11ad on a smartphone under Gbps data rates and compare it with the power consumption of 5 GHz technologies. We find that, surprisingly, the 802.11ad Tx power is significantly lower than the Rx power, and the lowest among the three technologies. Overall, we conclude that Gbps data rates introduced by 802.11ad do not have a significant impact on the smartphone's battery life.

In summary, our results show that 802.11ad has higher potential to address the needs of emerging bandwidth-intensive applications in smartphones than its 5 GHz counterparts. At the same time, we identify several open challenges and key research directions towards fully realizing this potential.

2. Related work

mmWave performance characterization. There is a large body of work on 60 GHz performance characterization using either SDRs [8,9,16,17,20–22] or WiGig-based hardware [2,10,23,24], which are not fully 802.11ad-compliant. In the case of SDRs, there are often additional issues such as the use of horn antennas instead of antenna arrays and the limited bandwidth [8,20,21]. More recent works [11–14,25–30]) have used 802.11ad-compliant COTS APs and laptops to study 802.11ad link characteristics or upper layer performance over 802.11ad links. None of the above works has conducted experiments with smartphones. The use of laptops as clients does not allow for performance evaluation under realistic mobility patterns. In contrast, our work is the first to evaluate 802.11ad performance on a real smartphone under typical scenarios that a smartphone user encounters in day-to-day operation.

Unlike performance, the power consumption of mmWave radios has not been studied extensively. There are only two works [2,11] that studied experimentally the power consumption of WiGig/802.11ad PCIE/M2 radios. In contrast, in this work, we evaluate the 802.11ad power consumption on a mobile device.

mmWave in smartphones. In our preliminary work [31], we conducted the first study of the performance of 802.11ad on a smartphone. This work extends [31] in the following ways: (i) We perform a much more detailed performance characterization, looking at the interplay between rate adaptation and beamforming and their impact on performance in various scenarios. (ii) We study for first time the impact of blockage in 802.11ad AP-smartphone links. (iii) We compare the performance and power consumption of 802.11ad on smartphones against that of 802.11ac and, for first time, 802.11ax. Recently, Narayanan et al. [32] conducted the first measurement study of mmWave 5G using smartphones equipped with 5G NR radios. Since the characteristics of cellular networks are very different from those of WLANs, this study is orthogonal to ours, focusing on topics such as comparison among different carriers, 5G–4G handoffs, location-based performance estimation, and impact on HTTP/2 and HTTP/3 performance.

3. Experimental methodology

3.1. Devices

Since there is no device on the market featuring both an 802.11ad and an 802.11ax radio, we use different router–phone pairs for experimenting with different technologies.

3.1.1. 802.11ad setup

We use an ASUS ROG Phone and a Netgear Nighthawk X10 Smart WiFi Router.

The ASUS ROG Phone was the first commercially available smartphone with an 802.11ad chipset. It is powered by a Snapdragon 845 octa-core processor (max CPU frequency 2.95 GHz) with 8 GB RAM and a 4000 mAh battery. It supports all the 802.11ad single-carrier (SC) MCSes (1–12), which correspond to PHY data rates from 385 Mbps up to 4.6 Gbps. Its phased array consists of 8 elements and is located at the bottom left of the back of the device (Fig. 1). Note that the phone is marketed as a gamer's phone and the intended use of the 802.11ad chipset is to cast the phone's screen via an 802.11ad dock [33]. However, in this paper we evaluate it as a general purpose 802.11ad phone. For comparison, we also conducted a few measurements with

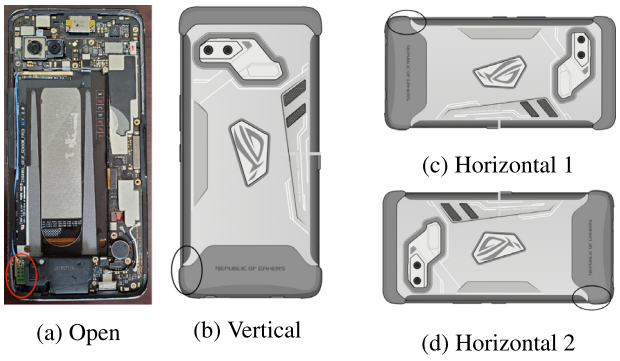


Fig. 1. Asus ROG Phone Orientations and Antenna Placement.

a ROG Phone II, which has the same WLAN features and CPU speed (although a newer CPU model) as the ROG model, but a larger battery (6000 mAh). Both devices yielded similar performance.

The Netgear Nighthawk X10 Smart WiFi router (one of the two commercially available 802.11ad routers) features a 32-element phased antenna array. The maximum throughput is limited to 2.3 Gbps when the client is placed right in front of it and to 1.65 Gbps in most practical scenarios [12,25,29,30].

3.1.2. 802.11ax setup

We used a Samsung Galaxy S10 and an ASUS RT-AX88U router.

The Samsung Galaxy S10 phone is powered by an Exynos 9820 octa-core processor (max CPU frequency 2.73 GHz) with 8 GB RAM and a 3400 mAh battery and houses an 802.11ax/ac chipset from Broadcom that supports all the 802.11ax MCSes (0–11), channel widths of 20/40/80 MHz, and 2 spatial streams, yielding PHY data rates from 6.5 Mbps up to 1.2 Gbps (866 Mbps) for 802.11ax (802.11ac). For comparison, we also conducted a few measurements with a Xiaomi Mi 10 phone, which is powered by a Snapdragon 865 octa-core processor (max CPU frequency 2.84 GHz) with 8 GB RAM and a 4780 mAh battery, and houses an 802.11ax/ac chipset from Qualcomm that supports exactly the same features as the S10's chipset. Both devices yielded similar performance.

The ASUS RT-AX88U router houses a Broadcom BCM43684 chipset which supports all 802.11ax MCSes, channel widths up to 160 MHz, 4 spatial streams, and MU-MIMO. However, the limited capabilities of the phones (2 spatial streams, 80 MHz) limit the maximum PHY data rate to those devices to 1.2 Gbps.

3.1.3. 802.11ac setup

We primarily used the Netgear Nighthawk router with the ROG phone in the performance evaluation experiments for a direct comparison against 802.11ad. The ROG phone houses an 802.11ac chipset from Qualcomm, which supports all the 802.11ac MCSes (0–9), channel widths of 20/40/80 MHz, and 2 spatial streams, yielding PHY data rates from 6.5 Mbps up to 866 Mbps. The router also includes an 802.11ac chipset, which supports channel widths up to 160 MHz, 4 spatial streams, and MU-MIMO. However, the limited capabilities of the phones, as described above, limit the maximum PHY data rate to the phones to 866 Mbps.

3.2. Methodology

We used TCP in our evaluation, as most popular applications today run over TCP. We used iperf3 to generate TCP traffic and logged throughput every 100 ms. In the range experiments (Section 4), we collected one 60 s trace and present the mean and standard deviation (error bars) of all the samples in the trace. In all other cases, we collected 5 traces of 30–60 s each and present their average and standard deviation. In the 802.11ad experiments, we logged the Tx and Rx MCS and beamforming sectors used by the phone and the router every 100 ms, using the information provided by the `wil6210` driver on the phone. In the 802.11ac and 802.11ax experiments, since the drivers do not export any information, we set up a laptop in monitor mode and captured MAC headers from which we extracted the selected MCS, number of spatial streams, and channel width. All the experiments were performed at night to avoid interference in the 5 GHz band from the campus networks and human blockage in the 60 GHz band.

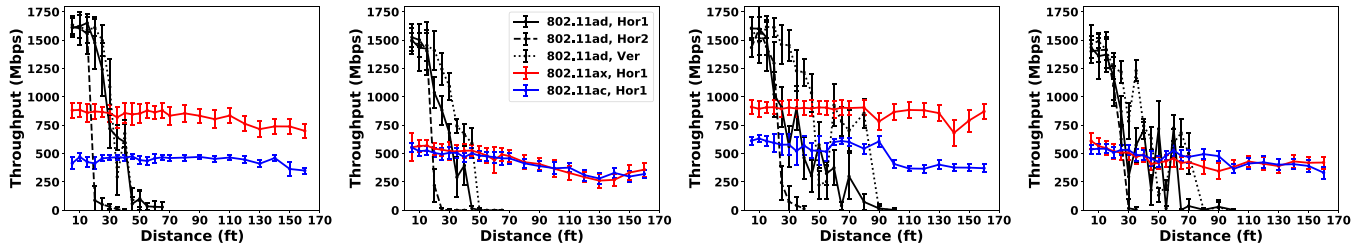
4. Performance in static settings

4.1. Baseline comparison and range

In our previous work [12], using off-the-shelf 802.11ad routers (including the Nighthawk router we use in this paper) and laptops, we showed that, in contrast to common belief, 802.11ad achieves excellent indoor range, with Gbps speeds often supported at distances longer than 100 ft. In this section, we examine if an 802.11ad-enabled smartphone can support similar ranges. We performed measurements in the same two locations described in [12], for a direct comparison between the laptop and the phone: a wide open space (lobby) with a high ceiling and a few desks and chairs, and a long, narrow (5 ft) corridor, completely free of any objects, with dry wall on both sides. In both locations, we kept the AP at a height of 6 ft and the phone on a stand at a height of 3.5 ft with the back of the phone facing the AP, and performed measurements in both the uplink (phone to AP) and downlink (AP to phone) direction. We repeated the measurements with three different phone orientations (Fig. 1), since the device orientation affects the position of the phone's antenna array with respect to the AP and hence, the beamforming sector selection. We found that the impact of orientation is minimal on the performance of 802.11ac/ax, so we only show results with one orientation (Horizontal 1), which yielded slightly better performance than the other two, for the two 5 GHz technologies.

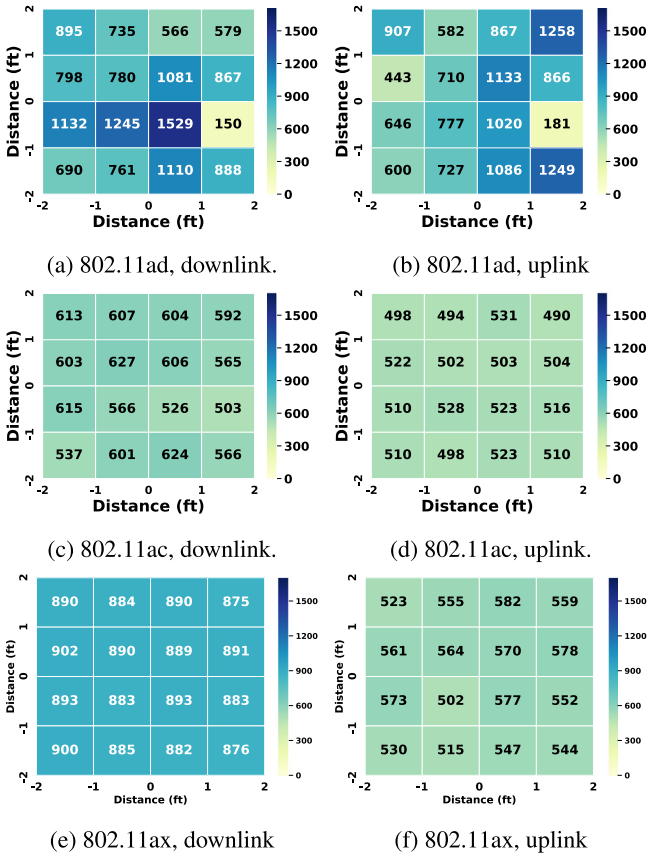
Figs. 2(a)–2(d) show that, although the range of AP-phone links is much lower than the range of AP-laptop links, *Gbps speeds are still feasible with the phone at distances of several tens of ft in both directions*. Overall, the Horizontal 1 orientation achieves a longer connectivity range than the Vertical orientation, but the Vertical orientation yields higher throughput. The Horizontal 2 orientation performs much worse than the other two. This is interesting, given that the phone is marketed as a gaming phone and one would consider a horizontal orientation to be the optimal one. Another observation from these figures is that, in the lobby, the throughput exhibits a steep drop; in contrast, in the corridor, the range is much longer but the throughput with the Horizontal 1 and Vertical orientations exhibits large spatial variations even at positions 5 ft apart, due to the waveguide effects (also reported in [12] for AP-laptop links). Note that in both locations, the max 802.11ad throughput is limited to only 1.65 Gbps, much lower than the max supported throughput of 2.2 Gbps. This is because the highest supported MCS is 8 (see Fig. 6); MCSes higher than 8 are only supported at very short distances (1–2 ft) from the AP. The same observation was made for AP-laptop links in [12,13,29].

As expected, the two 5 GHz technologies have a much longer range, yielding throughput higher than 300 Mbps up to 170 ft and the standard deviations are much lower compared to 802.11ad. We also observe that the impact of location is much less pronounced at 5 GHz. 802.11ac yields a throughput of 400–600 Mbps in both directions at both locations. On the other hand, we observe a large disparity between the uplink and downlink throughput for 802.11ax, similar to 802.11ad; the former is about 300 Mbps lower and very similar to that of 802.11ac, although the same MCSes are used in both directions. Since both phones use 2 spatial streams and 80 MHz, the higher downlink throughput with 802.11ax compared to 802.11ac is due to the higher MCSes (10, 11) supported only by 802.11ax. Overall, we observe that the hardware limitations of this first generation of 802.11ax smartphones (without support for 160 MHz channels and 3 or 4 spatial streams) prevent them from truly achieving Gbps rates, *making 802.11ad the only true Gbps WLAN technology to date*.



(a) Lobby, Downlink. (b) Lobby, Uplink. (c) Corridor, Downlink. (d) Corridor, Uplink.

Fig. 2. Throughput over distance.



(a) 802.11ad, downlink. (b) 802.11ad, uplink

(c) 802.11ac, downlink. (d) 802.11ac, uplink.

(e) 802.11ax, downlink (f) 802.11ax, uplink

Fig. 3. Throughput (in Mbps) with each technology in a 16 ft^2 area, 30 ft away from the AP, in the corridor.

4.2. Spatial variations

We saw in Figs. 2(a)–2(d) that 802.11ad throughput can vary by several 100 s of Mbps at distances 5 ft apart, especially in the corridor. We further explore this behavior by measuring throughput in an area of 16 ft^2 divided into 16 grids, at different distances from the AP. Fig. 3 shows an example of such measurements. We observe that *a small displacement of the phone by 1 ft can result in a throughput reduction by an order of magnitude*. We confirmed this extreme sensitivity at multiple distances from the AP in both locations. *A direct implication of this finding is that coarse-grained throughput prediction, e.g., only based on the AP-client orientation [14] is not always feasible, especially in environments with multiple nearby reflectors*. In contrast, both 5 GHz technologies are totally insensitive to such small displacements (Figs. 3(c)–3(f)).

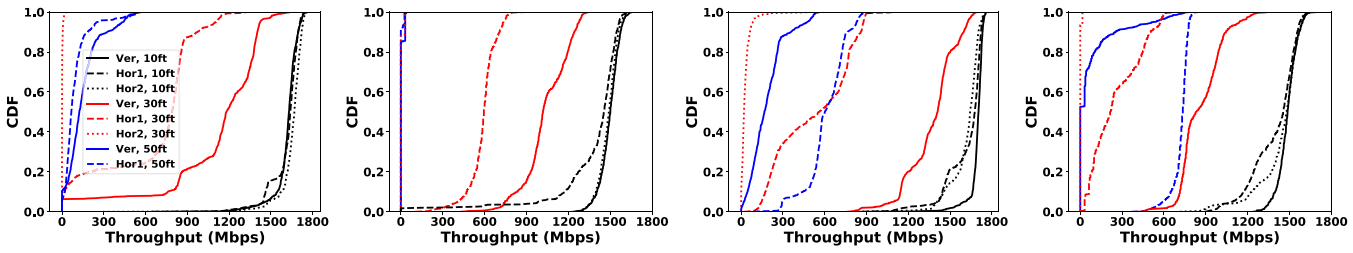
4.3. Temporal variations

We now take a closer look at temporal throughput variations by focusing on three representative AP-client distances (short, medium, long) for each technology: 10 ft, 30 ft, 50 ft for 802.11ad and 10 ft, 80 ft, 160 ft for 802.11ac/ax. Fig. 4 plots the CDFs of the 802.11ad throughput samples, taken over a 60 s period in both locations. We observe that *802.11ad throughput exhibits very large temporal variations*, especially at medium distances (30 ft), but also at long distances (50 ft) for some orientations and/or directions. As an example, the downlink throughput in the lobby at 30 ft for the Vertical Horizontal 1 orientations spans the whole range from 0 (10% of the time) up to 1.5 Gbps (Fig. 4(a)). As another example, the uplink throughput in the corridor at 150 ft for the Vertical orientation is 0 55% of the time and varies from 50–750 Mbps the rest of the time (Fig. 4(d)). On the other hand, at short distances (10 ft), throughput is much more stable but exhibits long tails (e.g., the uplink throughput for the Horizontal 1 orientation in the lobby, Fig. 4(b)). In contrast, the performance at 5 GHz is much more stable over time (Fig. 5); the variations are typically limited to a range of 100 Mbps. Still, there are a few exceptions, mostly for 802.11ax uplink, e.g., vertical orientation at 80 ft in the lobby (Fig. 5(b)), where throughput varies from 100–650 Mbps.

The large 802.11ad throughput variations are partly due to the fact that RA cannot lock at a single MCS; we observe (Fig. 6) 4–6 different MCSes at 30 (and sometimes 50) ft and the dominant MCS is often used for only 40%–60% of the time, especially in the downlink. In the uplink, the beamforming protocol, which aggressively triggers a sector level sweep (SLS) upon a missing link layer ACK, also contributes equally to the large throughput variations (Figs. 7(b), 7(d)). The phone selects up to 4–6 different Tx sectors within 60 s even at 10 ft, and up to 11 sectors at 30/50 ft. We also observe that in certain cases, the transmitting device failed to find a working sector (white color in Figs. 7(a)–7(d)) at 30 ft and 50 ft for a fraction of time, which is typically small (2%–3%) but it can reach up to 18% (Horizontal 2, Lobby, uplink, 30 ft). In contrast, the TX sector selection on the AP (Figs. 7(a), 7(c)) is very stable.

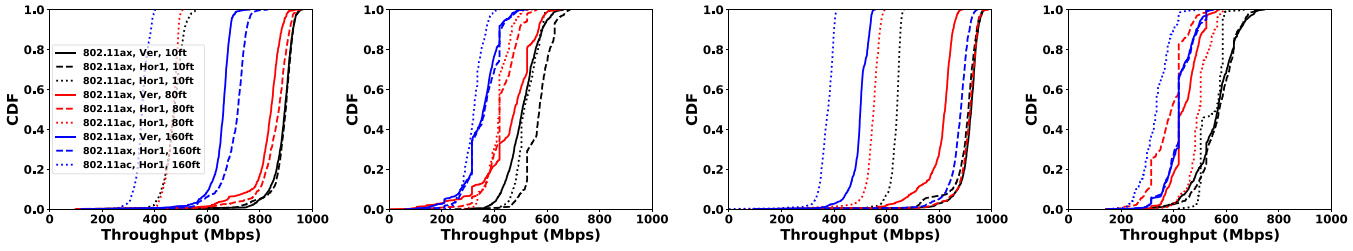
In case of the Rx sectors, we make a similar observation as in [12]: the Rx sector never changes. We draw the same conclusion here for the phone as in [12] for APs/laptops; the phone uses a quasi-omni beam pattern in Rx mode.

In contrast to 802.11ad, the RA logic makes much more stable decisions in the case of 802.11ac/ax, which contribute to the much smaller temporal throughput variations. Recall that the rate adaptation logic in 802.11ac/ax determines not only the MCS but also the number of spatial streams and the channel width. Fig. 8 shows that both protocols used almost exclusively 80 MHz channel width and 2 spatial streams, even at distances of 160 ft. In addition, in the downlink direction, we observe a dominant MCS at least 80% of the time in most cases, which is often among the top two (8/9 for 802.11ac, 10/11 for 802.11ax) even at 160 ft. Note that the first experimental study of 802.11ac performance back in 2014 [34] reported that MCS 8 and 9 were practically useless for distances longer than 30 ft; our results show that the technology has matured significantly over the past 6



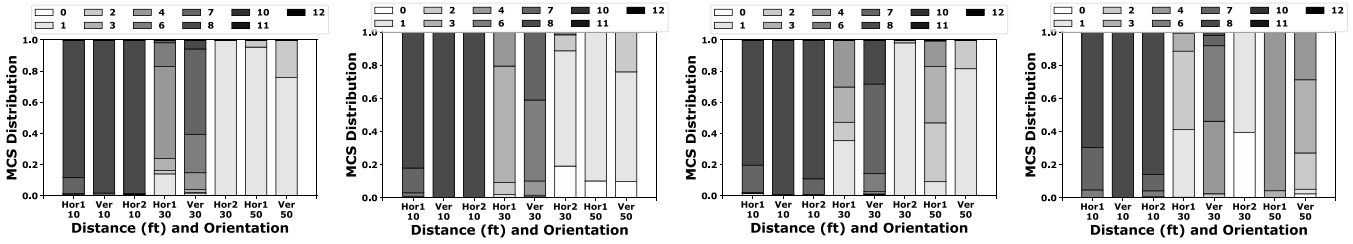
(a) Lobby, Downlink. (b) Lobby, Uplink. (c) Corridor, Downlink. (d) Corridor, Uplink.

Fig. 4. CDF of 802.11ad throughputs at 10, 30, and 50 ft.



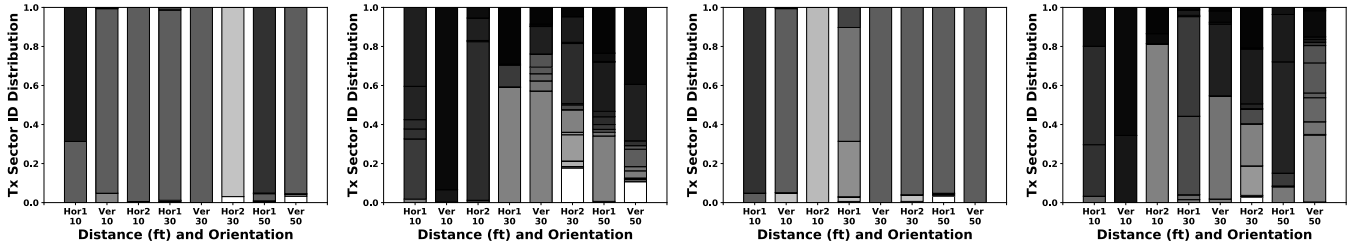
(a) Lobby, Downlink. (b) Lobby, Uplink. (c) Corridor, Downlink. (d) Corridor, Uplink.

Fig. 5. CDF of 802.11ac/ax throughputs at 10, 80, and 160 ft.



(a) Lobby, Downlink (b) Lobby, Uplink (c) Corridor, Downlink (d) Corridor, Uplink

Fig. 6. 802.11ad MCS distribution at 10, 30, and 50 ft. MCS 0 is used only for control frames.



(a) Lobby, Downlink (b) Lobby, Uplink (c) Corridor, Downlink (d) Corridor, Uplink

Fig. 7. 802.11ad Tx Sector ID distribution at 10, 30, and 50 ft. Different versions of gray denote different valid sector IDs. White color denotes the cases when beamforming did not find any working sector. Both the phone and the AP use a quasi-omni beam pattern in Rx mode.

years. In the uplink direction, the MCS selection is less stable (up to 8 different indexes); this explains the higher uplink throughput variations in Figs. 5(b), 5(d). We also observe that MCS 10–11 are only chosen at short distances (10 ft) but even at those distances, they cannot yield performance higher than that of MCS 9 (around 600 Mbps).

5. Coverage

In the previous section, we only evaluated the impact of distance between the phone and the AP. In this section, we consider both the distance and angular separation. We perform the experiments in 2 locations: the lobby and a lab. We divided the floor in both locations into 8 ft by 8 ft squares and took throughput measurements at the center of each square. Since the Horizontal 2 orientation yielded very poor performance in the previous experiments, in the following we only show experiments with the Vertical and Horizontal 1 orientations.

Figs. 9(a) and 9(b) plot the 802.11ad results for the lobby. We observe that the range changes significantly when angular separation is introduced. It drops sharply on the left side of the AP to 32 ft from 72 ft in front of the AP, while on the other side a connection can be sustained up to a much longer distance (56 ft) for both orientations. This is because of reflections from the wall on the right side of the lobby.

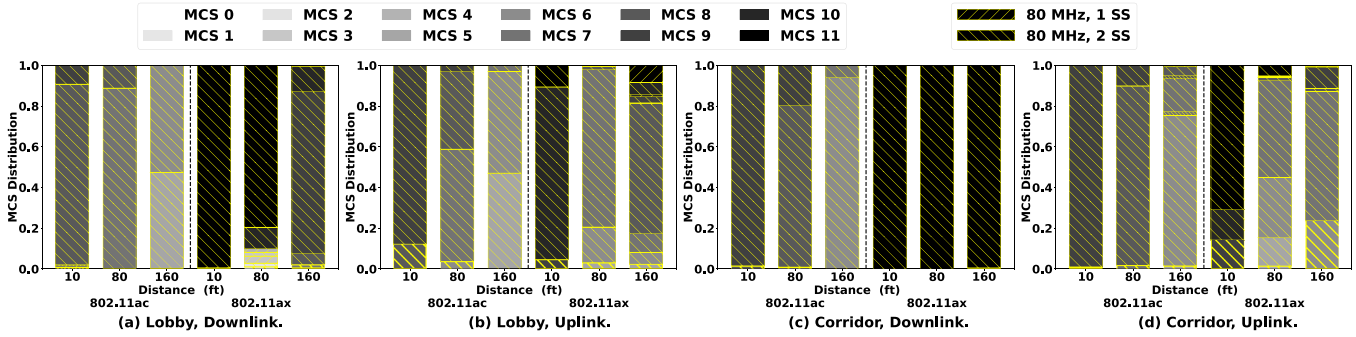


Fig. 8. 802.11ac/ax MCS distribution at 10, 80, and 160 ft.

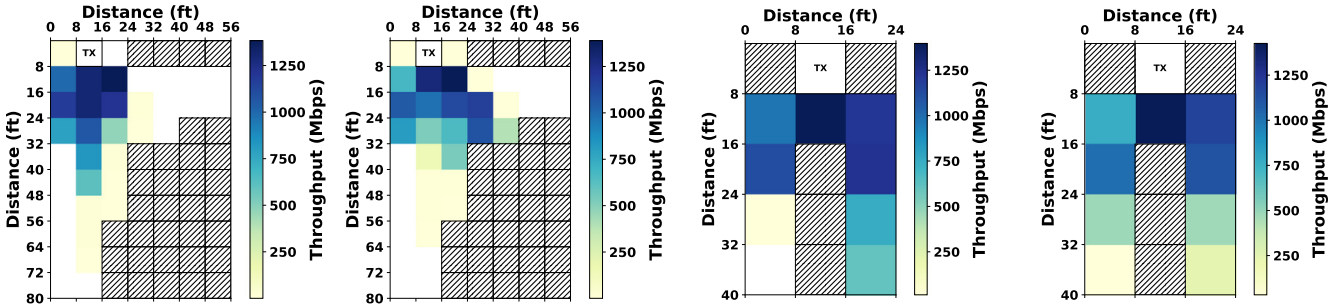


Fig. 9. 802.11ad coverage in Lobby and Lab with different orientations. The shaded region in the lobby shows inaccessible areas, such as walls and other rooms. The shaded region in the lab shows areas occupied by furniture.

Overall, coverage is much worse as compared to what was reported in [12] using a laptop, especially at narrower angles of separation. Additionally, coverage is much worse compared to that of 802.11ax and 802.11ac, although throughput at short distances is higher. We found that both 5 GHz technologies provide roughly uniform coverage throughout the entire lobby.

We also observe that each of the 2 orientations in the lobby works better in different positions. The Horizontal 1 orientation can establish a connection and achieve higher throughputs at wider angles while the Vertical orientation can achieve much higher throughputs at narrower angles. For example, the Vertical orientation can barely achieve 10–20 Mbps at the right-most position where it gets a connection whereas at the same position Horizontal 1 yields more than 1 Gbps throughput. On the other hand, at zero angular separation, the Vertical orientation achieves Gbps rates for longer distances than the Horizontal 1 orientation (40 ft vs. 24 ft).

The results in the lab are plotted in Figs. 9(c) and 9(d), again only for 802.11ad. Here, the performance is mostly symmetric with respect to location for the Horizontal 1 orientation, which is expected, since the lab environment is also symmetric. Surprisingly, for the Vertical orientation, the performance differs based on which side of the lab the phone is in. Note that because of the antenna placement on the phone, when the phone is on the right side of the AP, the antenna points towards the wall and the phone is able to sustain a connection and close to Gbps performance because of reflections from the wall. For example, for the furthest position on the left of the AP in Fig. 9(c), the phone cannot even establish a connection while on the other side it gets a throughput of ~600 Mbps. For the two sub-6 GHz technologies, the throughput values were within a few Mbps in all 9 grids.

Overall, *the 802.11ad coverage with a phone is lower compared to with a laptop and much worse compared to sub-6 GHz WiFi. Additionally, 802.11ad coverage is affected significantly by the device orientation, location, and AP-client angular separation. In contrast, 802.11ac and 802.11ax offer almost uniform coverage.*

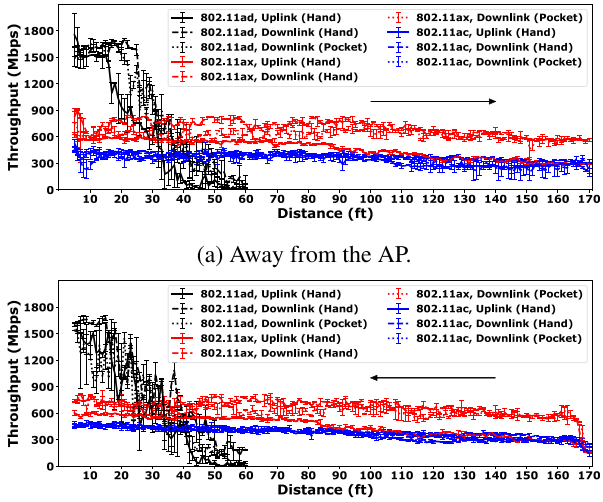
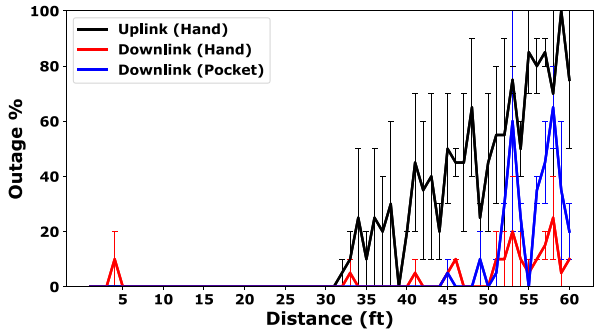


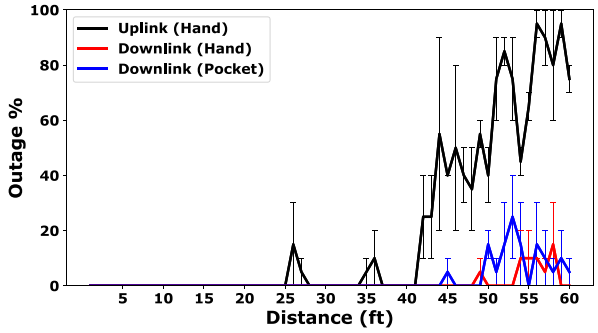
Fig. 10. Mobility in Lobby while facing the AP.

6. Mobility

In this section, *we evaluate the performance of 802.11ad under realistic smartphone mobility scenarios with the device in the user's hand or pocket.* This is the *first* study, to our best knowledge, that evaluates the performance of 802.11ad over realistic mobility patterns, as all previous works used either laptops (e.g., [10,12–14,23,30]) or bulky SDRs (e.g., [8,9]), which restricted motion.



(a) Away from the AP.



(b) Towards the AP.

Fig. 11. Outage time with mobility in Lobby while facing the AP.

6.1. Moving towards/away from the AP

We consider the following 3 scenarios: (a) moving away from the AP while facing away from the AP, (b) moving away from the AP while facing the AP, and (c) moving towards the AP, starting at a distance of 60 ft (802.11ad)/170 ft (802.11ac/ax). We repeat these measurements with the phone in the user's hand and in the user's pocket in the Vertical orientation. In all cases, the user moves at a speed of 1 ft/s.

When moving away from but facing towards the AP (Fig. 10(a)), 802.11ad sustains throughputs above 1.5 Gbps up to 15 ft when held in the hand, in both the uplink and downlink direction. After 15 ft, the uplink throughput drops very sharply. When moving towards the AP (Fig. 10(b)), both the uplink and downlink throughputs exhibit much higher variations over distance compared to the moving away case. Surprisingly, *keeping the phone in the pocket results in similar performance to holding it in the hand* for the whole duration of the motion in both directions. Figs. 10(a), 10(b) also show that 802.11ac/ax can sustain throughputs of several 100 s Mbps for distances up to 170 ft, although lower than in the static case (Section 4). Overall, mobility causes much smaller throughput variations in the 5 GHz band compared to the 60 GHz band.

We also plot the 802.11ad outage time, i.e., the percentage of time the throughput was zero at any given distance in Fig. 11 (for 802.11ac and 802.11ax, the outage time was zero in all 3 scenarios at all distances). When the user moves away from the AP, the downlink outage time when the phone is held in the hand is at most 25% for distances longer than 50 ft (Fig. 11(a)). When the phone is in the pocket, the average outage time goes up to 60% for similar distances. In contrast, the uplink outage time increases sharply to above 80% and exceeds 20% even for shorter distances of 30–40 ft. When the user moves towards the AP, the uplink outage times are similar but the downlink times are even lower compared to the moving away case — below 15% when the phone is in the hand and below 25% when it is in the pocket (Fig. 11(b)). As a comparison, the authors in [12] reported downlink outage times as high as 80% with the same Nighthawk router and a laptop in the lobby but at distances longer than 140 ft. On the other hand, they found no outage till 70 ft.

Fig. 12 plots representative timelines of the selected MCS and Tx sector ID when the user moves away from/towards the AP, while facing the AP. When the user moves away from the AP holding the phone in their hand, we observe three distinct phases in the downlink case (Fig. 12(a)). From 0–25 ft, the MCS is fixed at 8 and the AP tries a couple of sectors initially before it converges to sector 20; from 25–35 ft, the MCS varies mostly between 4–8 but the sector ID remains stable; finally, for distances longer than 35 ft, the AP keeps switching MCS (1–4) and starts trying multiple sectors (we also observe a few cases when the AP fails to find a working sector). When the phone is in the pocket (Fig. 12(b)), we still observe the same three phases, although both the MCS and sector variations are higher and start earlier, and the number of beamforming failures increases too.

Interestingly, in the uplink case (Fig. 12(c)), the behavior is different. The sector selection becomes very unstable; the phone keeps trying several sectors for the whole duration of the motion and the number of failures is even larger than in Fig. 12(b). On the other hand, the MCS selection exhibits much less variation but there are multiple instances where MCS drops to 0, i.e., data transmission in not feasible.

Finally, when the user moves towards the AP (Figs. 12(d)–12(f)), we observe larger variations of the selected MCS for each of the three cases compared to moving away, as the AP/phone aggressively tries higher MCSes [35,36], which explain the throughput variations in Figs. 10(a), 10(b).

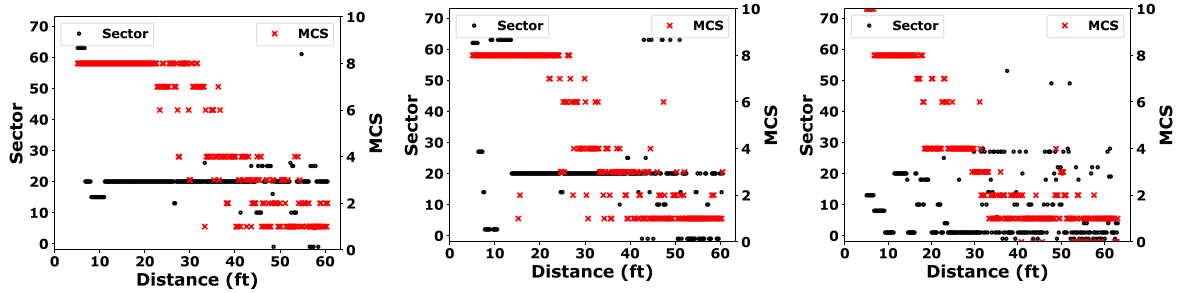
Note that in this scenario, where the user moves on a straight line always facing the AP, the same sector should always work (except for very short distances, due to the 3D shapes of the beams and the fact that the AP and the phone are not kept at the same height) and there should not be any need for beamforming; packet loss should be repaired by RA only. As we saw, this is in general the case on the AP for a large segment of the route (Figs. 12(a), 12(b) and 12(d), 12(e)), albeit increased packet loss due to reduced RSS at long distances or when the phone is in the pocket increases the number of unnecessary beamforming operations. On the other hand, this is clearly not the case on the phone (Figs. 12(c), 12(f)).

When we moved facing away from the AP, 802.11ad always lost connection within the first 5–10 s. In this case, the body completely blocked the signal between the AP and the phone and neither of them were able to find a working Tx sector even through reflections, except at very short distances. In contrast, the 802.11ac/ax performance was similar to that in Figs. 10(a), 10(b).

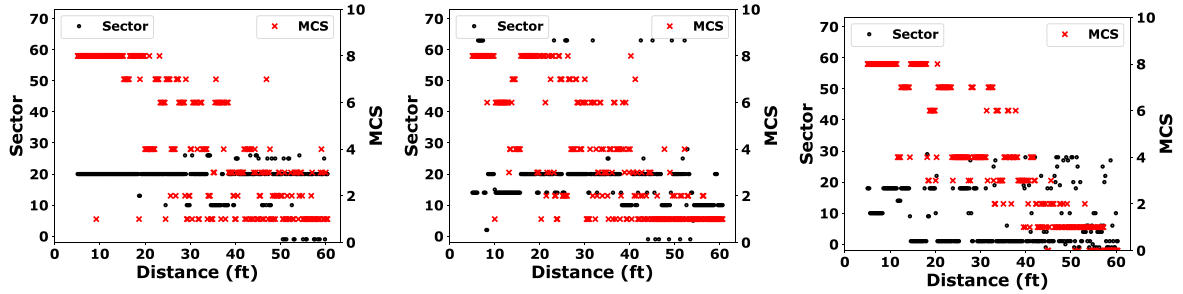
6.2. Lateral motion

We performed experiments while moving laterally w.r.t. the AP, eastbound and westbound, at a longitudinal distance of 5 ft and 15 ft from the AP. In all experiments, we started walking 30 ft away from the AP, passed in front of the AP, and continued for 30 ft more.

Fig. 13 shows that the 802.11ad range was much shorter compared to the static case in Fig. 2 and the moving towards/away case in Fig. 10, due to the combined effect of self-blockage, mobility, and antenna orientation. Note that the phone's antenna array pointed towards the AP when the phone moved eastbound but away from the AP when it moved westbound. As a result, the 802.11ad range of motion was longer and the throughput higher in the former case. In fact, at a longitudinal distance of 15 ft, while moving westbound (Fig. 13(d)), the phone never established a connection to the AP, as the combined effect of the body blockage and longer distance cause a much larger signal attenuation. On the other hand, while moving eastbound 15 ft away from the AP (Fig. 13(c)), the range of motion was significantly higher than in the other cases, as the angular separation for the same lateral distance was smaller due to the larger longitudinal distance from the AP. We also observe that the throughput often obtained its maximum value before the point in front of the AP, as once the user passed

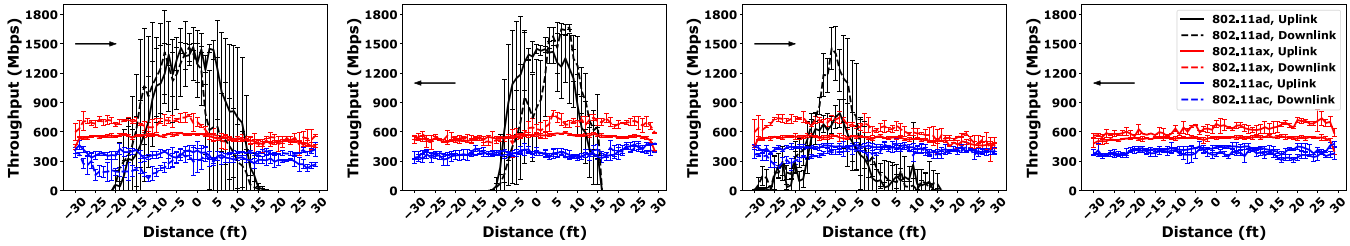


(a) Away, downlink, hand. (b) Away, downlink, pocket. (c) Away, uplink, hand.



(d) Towards, downlink, hand. (e) Towards, downlink, pocket. (f) Towards, uplink, hand.

Fig. 12. Mobility in Lobby while facing the AP: Timelines of MCS and Tx sector IDs. MCS 0 is used only for control frames. Sector ID -1 is reported when no working sector is found.



(a) Eastbound, 5 ft from AP. (b) Westbound, 5 ft from AP. (c) Eastbound, 15 ft from AP. (d) Westbound, 15 ft from AP.

Fig. 13. Lateral Mobility.

that point, their body blocked the signal, leading to repeated SLSs and unstable MCS selection. Interestingly, in this case, beamforming appears more effective on the phone than on the AP at a distance of 5 ft from the AP (Figs. 13(a), 13(b)), resulting in higher uplink than downlink throughput after crossing the point in front of the AP.

On the other hand, 802.11ac/ax maintained much more stable performance for the whole duration of the motion, although lower than in the static and the moving towards/away cases for the same distance. Still, we observe the impact of motion on RA decisions. For example, when moving eastbound (Figs. 13(a), 13(c)), the 802.11ax downlink throughput dropped by 200 Mbps when we passed in front of the AP and never recovered. In contrast, the 802.11ac throughput quickly drops down to 200–250 Mbps (5 ft after we start moving) but increases to 500 Mbps when we pass in front of the AP and remains at the same level even 30 ft away.

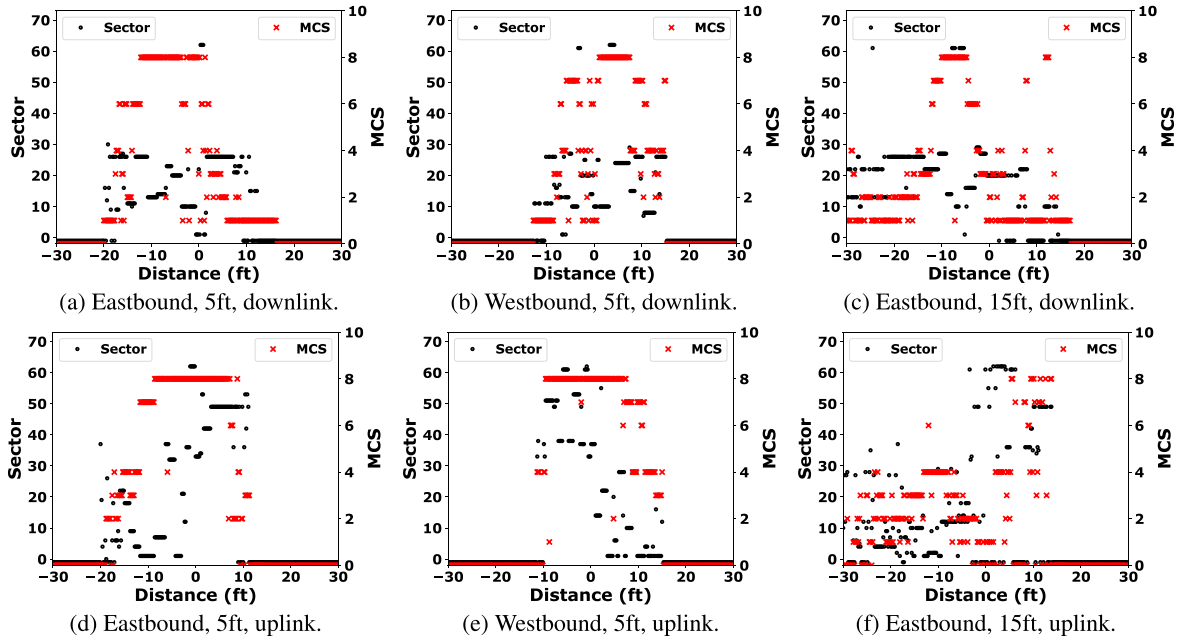
Fig. 14 plots representative timelines of the selected MCS and Tx sector ID when the user moves laterally w.r.t. the AP (Fig. 13). Figs. 14(a), 14(b), and 14(d), 14(e), explain the higher uplink performance compared to downlink at a longitudinal distance of 5 ft from the AP, which we observed in Figs. 13(a), 13(b). The AP in this case keeps trying different sectors (Figs. 14(a), 14(b)) but makes wrong decisions (e.g., in Fig. 14(a), it selects the same sector for segments of the path before and after the user crosses the point in front of the AP), which result in large MCS variations and poor downlink performance. On the other hand, the phone also tries multiple sectors (Figs. 14(e), 14(d)), but it makes a better selection, resulting in stable, high MCS for longer segments of the traversed path and higher uplink performance.

At a longitudinal distance of 15 ft from the AP, the sector selection on the AP (Fig. 14(c)) is much more stable than on the phone (Fig. 14(f)), resulting in higher performance, as we observed in Fig. 13(c). We also observe in this case that both the AP and the phone aggressively try high MCSes at the beginning of the route (15 ft to 0 ft), even though such MCSes cannot be sustained, causing packet losses and further throughput degradation.

Overall, the results in Section 6.1, Section 6.2 confirm our findings in Section 4 – *beamforming generally becomes much more challenging in small form factor devices*. Nonetheless, *802.11ad is able to sustain Gbps throughputs even under realistic mobility patterns, although the position of the antenna array with respect to the AP along with self-blockage greatly affect both the performance and the range of motion*.

6.3. Micro-mobility

Lastly, we evaluate 802.11ad performance in a micro-mobility [36] scenario. In this experiment, the user is sitting in a chair 10 ft from the AP playing a racing game on the phone, while at the same time we are running an iperf3 TCP session from the phone to the AP, emulating a scenario where the phone uses the 802.11ad link to cast the screen via a dock to a TV (the default usage scenario for the ROG phone). The phone initially is held in the Horizontal 1 orientation and, while the game is



(a) Eastbound, 5ft, downlink. (b) Westbound, 5ft, downlink. (c) Eastbound, 15ft, downlink.

(d) Eastbound, 5ft, uplink. (e) Westbound, 5ft, uplink. (f) Eastbound, 15ft, uplink.

Fig. 14. Lateral Mobility: Timelines of MCS and Tx sector IDs.

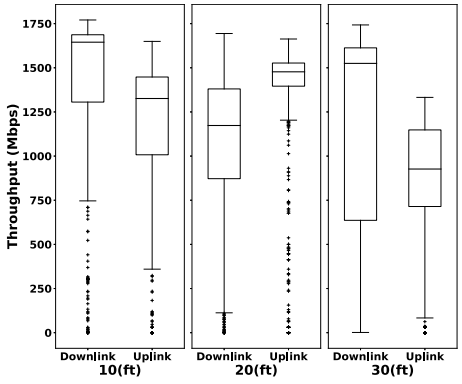


Fig. 15. Impact of mobile human blockage.

played, the user moves the device around in random positions, while playing the dame. The average throughput (100 ms samples) during 3 runs was 1.29, 1.38, and 1.33 Gbps with a standard deviation of 184, 161, and 284 Mbps. Overall, *802.11ad performs very well in its default usage scenario*.

7. Blockage

The work in [12] showed that the impact of human blockage in the case of COTS AP-laptop links is much smaller than what is reported in several other works. Here, we examine if the impact is similar in the case of smartphones. We consider two types of human blockage, similar to the work in [12]: (i) mobile and (ii) quasi-static. In both cases, we used the lobby setup described in Section 4 and three different AP-client distances: 10 ft, 20 ft, and 30 ft. We only show results with 802.11ad, since human blockage has no impact on the 5 GHz performance.

Mobile blockage. We initiate a 30-s TCP iperf3 session during which a human moves continuously in and out of the LOS between the AP and the smartphone, blocking the link at different random positions. Fig. 15 presents boxplots of the throughput in each scenario. Similar to the results reported in [12] for AP-laptop links, we observe that *mobile human blockage does not affect significantly the 802.11ad performance on the phone*. The median downlink (uplink) throughput over the duration of each experiment is 1.2–1.65 Gbps (0.9–1.4 Gbps) at different AP-phone distances.

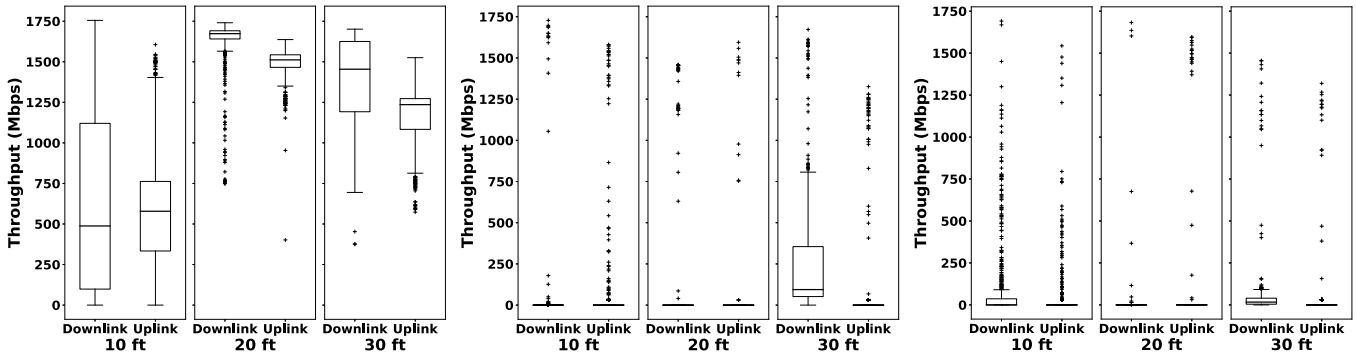
Quasi-static blockage. We initiate again a 30-s TCP iperf3 session. 5 s after the iperf3 session started, a human moves into the LOS between the AP and the phone, stands at the same position for 20 s, and then moves out. We consider three different blockage positions: near the AP, in the middle, and near the phone. Fig. 16 presents boxplots of the throughput during the 20 s of blockage in each scenario.

When the blockage is near the AP, the impact is significant only at 10 ft. For longer links, the impact is negligible, since the human is essentially only partially blocking the link. This result is similar to the result in [12]. However, our results are very different from those in [12] in the case of blockage in the middle of the link or near the phone; the throughput drops (very close) to 0 in all cases with one exception: blockage in the middle of a 30 ft link. In contrast, the work in [12] reported Gbps throughputs in both scenarios. As the human moves closer to the smartphone, blockage becomes stronger, and the poor beamforming capabilities of the phone compared to a laptop combined with the reduced range (see Section 4.1), result in extremely poor performance.

8. Power consumption

We conduct a preliminary evaluation of the power consumption for each of the 3 technologies. For 802.11ad and 802.111ac, we use the ROG phone, for a direct comparison between the two technologies. For 802.11ax, we use the Mi 10 phone, whose power consumption is much lower compared to the S10 phone. We log the current and voltage values from the `/sys/class/power_supply/battery` directory on the phone. We report the power consumption values after taking out the phone's base power consumption. For 802.11ad, we consider two orientations (Vertical and Horizontal 1) at 3 different distances (10, 30, and 50 ft). For 802.11ax, the power consumption is the same with both orientations at all 3 distances. The results are shown in Table 1.

At 10 ft, the average uplink/downlink 802.11ad power consumption is 1194 mW/1948 mW for the Vertical orientation, with uplink/downlink throughputs of 1.45 Gbps and 1.56 Gbps. We see similar values for the Horizontal 1 orientation. At 30 ft, the average uplink/downlink power consumption is 1452 mW/2174 mW for the



(a) Near the AP. (b) Middle. (c) Near the phone.

Fig. 16. Impact of quasi-static human blockage.

Table 1
Uplink (UL) and downlink (DL) power consumption (mW) with each technology.

802.11ad												802.11ac		802.11ax	
10 ft				30 ft				50 ft							
Vertical		Horizontal 1		Vertical		Horizontal 1		Vertical		Horizontal 1					
UL	*DL*	*UL*	*DL*	*UL*	*DL*	*UL*	*DL*	*UL*	*DL*	*UL*	*DL*	*UL*	*DL*	*UL*	*DL*
1194	1948	1294	1903	1452	2174	1113	1874	765	1409	670	1238	2886	1181	2058	1092

Vertical orientation, with uplink/downlink throughputs of 1.05 Gbps and 1.4 Gbps, and 1113 mW/1872 mW for the Horizontal 1 orientation, with uplink/downlink throughputs of 702 Mbps and 822 Mbps. We conjecture that the higher power values at 30 ft compared to 10 ft for the Vertical orientation, in spite of lower or comparable throughputs, are due to a higher number of beamforming operations [11]. Finally, at 50 ft, the power consumption is much lower with both orientations, since the throughput is also very low (a few tens of Mbps). At all three distances, we surprisingly see that *the 802.11ad power consumption for data reception (downlink) is much higher than for data transmission (uplink)*. Although we do not have an explanation, we note that a similar observation was made in [11] using a laptop. Table 1 also shows that the 802.11ac/802.11ax uplink power consumption is 2800 mW/2100 mW, making *802.11ad the most power-efficient technology*. On the other hand, in the downlink the power consumption is around 1100 mW with both technologies, much lower than the 802.11ad power consumption.

We also perform an experiment where we charge the battery to 100% and run a 20 min long data transfer in both the uplink and downlink directions separately. The battery percentage when the transfers are performed over 802.11ad drops by only 5%/7% in the uplink/downlink case. The numbers are similar for the two 5 GHz technologies: 9%/4% for 802.11ac and 6%/3% with 802.11ax. Overall, *Gbps data rates introduced by 802.11ad do not have a significant impact on the smartphone's battery life*.

9. Discussion

Our results in the previous sections showed that 802.11ad has higher potential to address the needs of emerging bandwidth-intensive applications in smartphones compared to its 5 GHz counterparts. At the same time, we identified several open challenges and key research directions towards fully realizing this potential, which we discuss below.

9.1. Open challenges

Beamforming in small form factor devices. Our results in Sections 4, 5, 6 indicate that beamforming is much more challenging in small form factor mobile devices compared to in laptops/APs; multiple beamforming events are triggered not only under mobility, but also in static scenarios. While the duration of an SLS is very short (lower than 1 ms) in this first generation of 802.11ad-enabled COTS devices, the large number of beamforming events and the poor interaction between beamforming and RA often results in large temporal and spatial performance variations and poor performance predictability not only under mobility, but also in static scenarios. In addition, poor beamforming capabilities lead to reduced coverage compared to the coverage achieved by 802.11ad laptops.

Antenna array placement. Our results in Section 4.1 showed very different performance and coverage for different ways of holding the phone, suggesting that the antenna array placement has a large impact on the 802.11ad performance and coverage in the case of small form factor mobile devices. In addition, their small form factor makes 802.11ad-enabled smartphones prone to self-blockage under different mobility patterns, as we showed in Section 6. Given that users might hold their phones in more than one ways, e.g., depending on the application, optimal antenna array placement in 802.11ad-enabled mobile devices remains an open challenge.

Sensitivity to human blockage. Our results in Section 7 showed that quasi-static human blockage can have a severe impact on the performance of 802.11ad-enabled smartphones. Since blockage can be unavoidable in certain crowded indoor WLAN environments and beamforming alone is not always able to recover from blockage events, alternative approaches are needed to limit the impact of blockage and/or ensure seamless connectivity even in the presence of blockage events.

9.2. Future research directions

Motion-aware link adaptation. Our results in Section 6 show the need for *motion-aware* beamforming and RA algorithms, which can (i) make decisions on when to trigger link adaptation, (ii) select which type of adaptation to trigger (beamforming or RA), and (iii) make more fine-grained decisions, e.g., select which sectors or MCSs to probe, based on the type of motion. A promising approach towards the design of such algorithms is by leveraging PHY layer information. Past research has successfully leveraged CSI and time of flight to design motion-aware RA algorithms for legacy WiFi [36]. In our recent work [37], we showed for the first time how to leverage PHY layer information

and machine learning to guide the selection of the right adaptation mechanism in 802.11ad WLANs. Nonetheless, there is still a need for fine-grained motion-aware RA or beamforming algorithms that will reduce the probing overhead under various mobility scenarios. An alternative approach is to leverage motion sensors; this approach might in fact be better suited for mobile devices due to the availability of motion sensors on such devices. The work in [35] has successfully demonstrated the efficacy of this approach in the case of legacy WiFi. On the other hand, the work in [38] proposed motion-aware beamforming for 802.11ad-enabled mobile devices, but the approach was only evaluated in simulations.

Use of arrays of phased arrays. The poor beamforming capabilities of small form factor mobile devices and the large impact of the antenna array position on performance calls for alternative approaches to improving coverage and resilience to self-blockage. One promising approach is the use of an *array of phased arrays*, as demonstrated recently in [39,40]. The work in [40] recently demonstrated this approach in the case of 802.11ad APs. In the case of small mobile devices, there are additional challenges. For example, the phased arrays might have to be distributed on difference faces of the mobile device, as proposed in [39] for an array of light sensors, to deal with the self-blockage problem. In addition, power consumption might be an issue for mobile devices. While our preliminary results in Section 8 showed that Gbps data rates introduced by 802.11ad do not have a significant impact on the smartphone's battery life, the downlink 802.11ad power consumption is much higher than the power consumption of legacy WiFi radios, and it may become prohibitive in the case of multiple phased arrays.

Bundling 5 GHz and 60 GHz interfaces. Another interesting approach is the bundling of 60 GHz and 5 GHz interfaces [29,30] to simultaneously offer high throughput in cases of LOS links and resilience to blockage or high mobility. In fact, the 802.11ad standard supports a Fast Session Transfer (FST) feature [1] to enable seamless migration to legacy WiFi when the 802.11ad link breaks. However, this feature is optional and is not implemented by any of the COTS devices available on the market. The work in [25] showed that combining legacy WiFi and 60 GHz radios using a higher layer solution (the Linux bonding driver) often leads to lower performance than using the 60 GHz radio alone, due to poor interaction with the TCP protocol that results in a large number of out of order packets, and concluded that legacy WiFi and 60 GHz radios should not be used simultaneously. More recently, the work in [30] demonstrated that it is possible to successfully bundle 60 GHz and legacy WiFi radios by modifying the Linux bonding driver for a specific application (video streaming). In our recent work [29], we demonstrated an alternative approach of bundling the two interfaces via Multipath TCP (MPTCP).

Nonetheless, the works in [25,29,30] have only been implemented in 60 GHz laptops. In the case of smartphones, there are additional challenges related to resource and power consumption. Note that the unique power features of 60 GHz radios compared to legacy WiFi radios (lower Tx power, higher Rx power) and the availability of both uplink-oriented (miracast, AR) and downlink-oriented (VR, video streaming) bandwidth-hungry applications makes the problem of designing power-aware bundling solutions for 60 GHz and legacy WiFi radios very different from the problem of bundling WiFi and cellular radios (e.g., in [41]).

10. Conclusion

We presented the first extensive measurement study of 802.11ad on commercial smartphones and compared for first time the performance and power consumption of 802.11ad against those of two 5 GHz WiFi technologies: the state-of-the-art 802.11ac and the upcoming 802.11ax. We showed that, although the performance, range, and coverage of 802.11ad are inferior on the phone compared to an 802.11ad-enabled laptop, Gbps data rates are still achievable with the phone up to a few tens of ft. Additionally, we found that the hardware limitations of this first generation of 802.11ax smartphones prevent them from truly achieving Gbps rates, making 802.11ad the only true Gbps WLAN technology. Overall, our results showed that 802.11ad has higher potential to address the needs of emerging bandwidth-intensive applications in smartphones than its 5 GHz counterparts. At the same time, we identified several open challenges – beamforming in small form factor devices, antenna array placement, sensitivity to human blockage – and key research directions – motion-aware beamforming and rate adaptation design, use of arrays of phased arrays for better coverage and resilience to self-blockage, bundling 5 GHz and 60 GHz interfaces – towards fully realizing this potential.

CRediT authorship contribution statement

Shivang Aggarwal: Conceptualization, Methodology, Investigation, Data curation. **Moinak Ghoshal:** Methodology, Investigation, Data curation. **Piyali Banerjee:** Methodology, Investigation, Data curation. **Dimitrios Koutsonikolas:** Conceptualization, Methodology, Writing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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